Reproducing a CFD-DEM Study of a Geldart-D Fluidized Bed in Python

One-Way vs Quasi-Two-Way Gas Coupling, compared with Liu & van Wachem (2019)

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Aim & Objectives

Aim

Reproduce the results of Liu & van Wachem (2019) for the NETL small-scale challenge bed with a DEM code written from scratch in Python.

Objectives

- Build a soft-sphere DEM solver (Numba) for 3,380 nylon beads, d_p = 3.256 mm.
- Couple it to the gas (Gidaspow drag) one-way and quasi-two-way.
- Reproduce Figs 1–5: drag models, ΔP–u_g and u_mf, ΔP time series, PSD, snapshots.
- Measure bed-to-bed scatter (14 beds per coupling) and find where the simple gas model fails.

Methodology

Particles (DEM, paper Table 1)

- Spring–dashpot contacts, k_n = 3654 N/m, e_n = 0.84, μ = 0.35; Δt = t_c/50 = 3.3 μs.
- 2-D monolayer, W = 0.23 m, 24 × 56 fluid cells (3 d_p); fluid update every 5 DEM steps.

Gas-solid coupling

- Drag per particle: F_d = β (u_g − v_p) V_p / ε_s, β = Gidaspow (Ergun ε < 0.8, Wen–Yu above).

one-way: $u_g = U_g / \varepsilon_g$

quasi-two-way: $u_g = (U_g - q_s) / \varepsilon_g$

q_s = row-averaged solids volume flux

- ΔP_bed = Σ F_d / (W d_p); bed weight / area = 905 Pa.
- Ramp U_g at ±0.2 m/s² (→ u_mf); hold 2, 3, 4 u_mf for 15 s, statistics 5–15 s at 50 Hz.

How Python Reproduces It

params.py
all Table 1–2 inputs, derived t_c, Δt, grid

↓

bed.py · run_settle.py
lattice fill, gravity settling

↓

sim.py @njit
cell list → contacts → walls → drag → integrate

↓

drag.py @njit
voidage, Gidaspow β, one-/two-way drag

↓

run_ramp.py · run_hold.py
ramp for u_mf; 15 s holds, 50 Hz output

↓

analysis.py · paper_figs.py
u_mf fit, PSD (SciPy), figures (Matplotlib)

- Numba JIT compiles the Python loops: ≈ 2.5 ms per DEM step for 3,380 particles.
- 28 ramp runs (14 beds × 2 couplings) + 6 hold runs, ≈ 3 h each.

RESULTS — reference paper (left) vs this work (right)

Gas momentum conservation per fluid cell

Liu & van Wachem (2019)

Gas momentum, solved in every fluid cell:

$$\frac{\partial(\varepsilon_g \rho_g \mathbf{u}_g)}{\partial t} + \nabla\cdot(\varepsilon_g \rho_g \mathbf{u}_g \mathbf{u}_g) = -\varepsilon_g \nabla p + \nabla\cdot(\varepsilon_g \boldsymbol{\tau}_g) + \varepsilon_g \rho_g \mathbf{g} + \mathbf{S}_p$$

$$\mathbf{S}_p = -\frac{1}{V_{cell}} \sum_{i \in cell} \mathbf{F}_{d,i}$$ (drag on the gas)

Each cell has its own ∇p (x and y), so the drag in every cell is balanced locally:

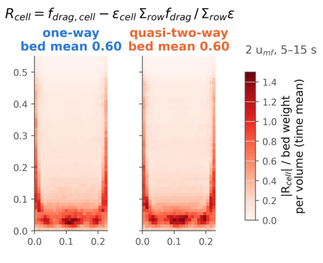
$R_{cell} = 0$ **in every cell**

→ gas turns round dense regions → bubbles

This work

No gas momentum eq.; one $\partial p/\partial y$ per row, $\partial p/\partial x = 0$:

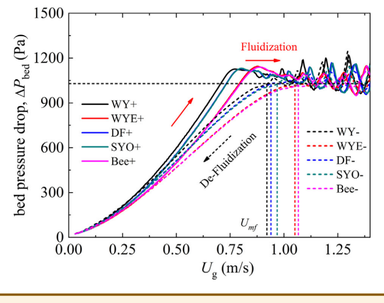
$R_{cell} = f_{drag,cell} - \varepsilon_{cell} \Sigma_{row} f_{drag} / \Sigma_{row} \varepsilon$



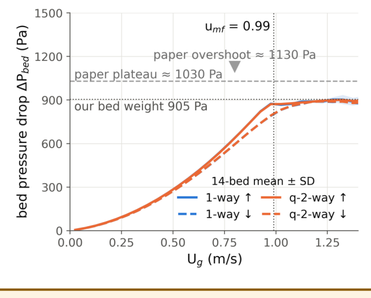
✘ Not conserved. Inside the bed, drag equal to 60 % of the local bed weight is left unbalanced by any pressure gradient (red). The paper balances every cell, so its gas can turn round dense regions and form bubbles; ours cannot.

Fig. 2 — ΔP_bed vs u_g (same axes)

Liu & van Wachem (2019)



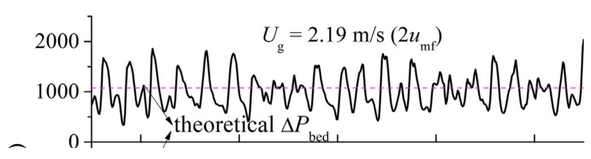
This work



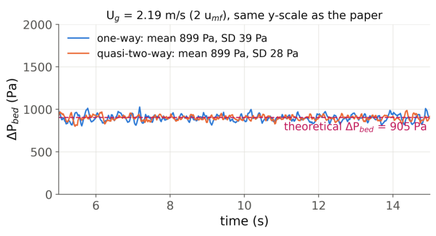
◐ u_mf matches (0.99 vs 1.00 paper, 1.05 expt). Missing: the 10 % overshoot at 0.8 m/s, the early rise (260 vs ≈ 550 Pa at 0.5 m/s) and the ≈ 100 Pa hysteresis. Lower plateau = lighter 2-D bed (905 vs 1075 Pa).

Fig. 3 — ΔP_bed time series at 2 u_mf

Liu & van Wachem (2019)



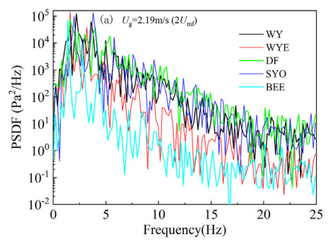
This work



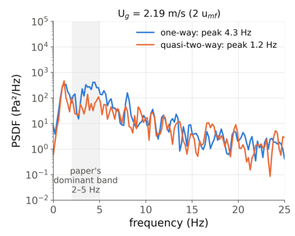
◐ Mean ΔP = 0.99 × weight (paper 0.94 ×) ✔. Fluctuations (SD ≈ 30–40 Pa) are much smaller than the paper's (≈ 500–1800 Pa swings).

Fig. 4 — Power spectrum of ΔP_bed at 2 u_mf

Liu & van Wachem (2019)



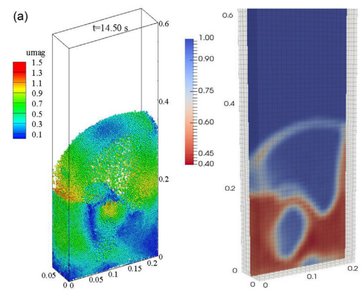
This work



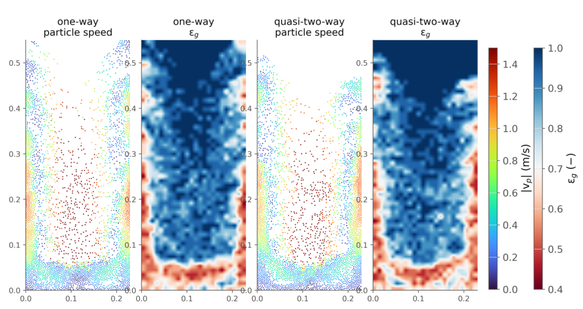
✔ One-way peak 4.5 Hz lies in the paper's 2–5 Hz band; quasi-two-way peak 1.2 Hz.

Fig. 5 — Snapshot at 2 u_mf, t = 14.5 s

Liu & van Wachem (2019)

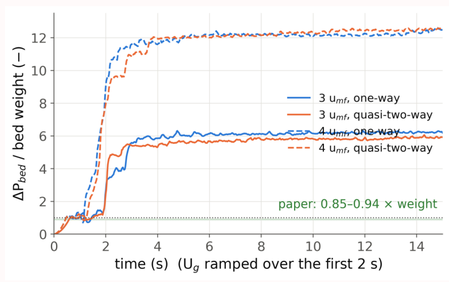


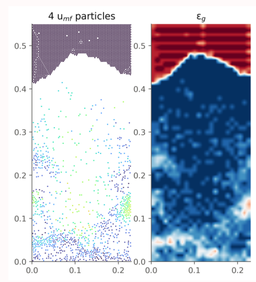
This work



◐ Bed fluidizes with a dense base and dilute core, but forms one large central void with solids rising at the walls; the paper shows discrete bubbles.

The Problem at 3 u_mf and 4 u_mf (this work)





- ΔP climbs to 6× (3 u_mf) and 12× (4 u_mf) the bed weight; the paper stays at 0.85–0.94×.
- Cause: the gas is solved only in 1-D (per row), so it cannot bypass the bed as bubbles. The whole bed lifts as a plug and is pressed against the top wall, in both couplings.

Summary of Numbers

Quantity	Paper	1-way	q-2-way
u_mf (m/s)	1.00	0.991	0.990
ΔP/W at 2 u_mf	0.94	0.99	0.99
ΔP/W at 3 u_mf	0.93	6.2	5.8
ΔP/W at 4 u_mf	0.85	12.2	12.3
PSD peak, 2 u_mf	2–5 Hz	4.5 Hz	1.2 Hz

W = bed weight / area (paper 1075 Pa, ours 905 Pa).
u_mf: mean of 14 beds per coupling.

Conclusions

- Drag closures, the ΔP–u_g curve and u_mf are reproduced within 1 % of the paper.
- At 2 u_mf the mean ΔP and the one-way ΔP frequency match; bubble shapes and ΔP swings do not yet.
- Quasi-two-way vs one-way changes u_mf by < 0.5 %: solids flux matters little near u_mf.
- Above 2 u_mf a 1-D gas field cannot form bubbles, so the bed plugs: the limit of this coupling.

Next Steps & References

- Solve the gas pressure in 2-D (porous-media/Darcy–Ergun) so gas can form bubbles at 3–4 u_mf.
- Lethe CFD-DEM (VANS) runs as a full two-way check (5 beds done; drag convention patched).
- Compare with NETL SSCP experiments (Gopalan et al., 2016) once the data are obtained.

[1] Liu, D., van Wachem, B. (2019) Powder Technol. 343, 145–158.
[2] Gopalan, B. et al. (2016) Powder Technol. 291, 299–310.
[3] Gidaspow, D. (1994) Multiphase Flow and Fluidization.
[4] Code: github.com/kaustubh2005ashok-tech/en649-fluidized-bed